

Milestone 2 implementation status

Customer segmentation, purchasing behaviour, engagement, revenue forecasting, personal forecasting and product-demand forecasting

Evaluation position	Functional Milestone 2 scope complete
Branch	Garvitk001
Last technical review	12 August 2026
Backend tests	17 passed
ML/preprocessing tests	10 passed
Frontend	Lint and production build passed

Current conclusion. The Milestone 2 pipeline is connected end to end: source data is processed into versioned model outputs, imported into the database, served by scoped FastAPI endpoints and displayed in the four React workspaces. Remaining items are data-provider confirmation and production hardening, not missing Milestone 2 pages.

1. Completed Milestone 2 work

Area	Implemented result	Evidence
Customer intelligence	Purchasing features, engagement score, K-Means and Hierarchical comparison, three business-readable segments.	5,878 customers; Silhouette 0.363917.
Business revenue	Daily INR revenue forecasts with 7/14/30-day horizons, category filters, intervals and candidate metrics.	forecast-v2; Linear Trend selected.
Personal sales	Separate seller-scale forecast from authorised cleaned sales records.	personal-forecast-v2; seller-scoped API.
Product demand	Store/product forecasts using lag, calendar, discount, activity, stock and weather features.	forecast-v1; XGBoost; R2 0.918.
Backend and database	Versioned model runs, predictions, jobs, idempotent imports and Alembic schema.	Ruff, 17 tests and Alembic check passed.
Frontend	Real reports, charts, filters, search, pagination, exports and loading/error states.	Lint and Vite production build passed.

Revenue model improvement

The new weekday-aware Linear Trend candidate reduced aggregate revenue MAE from 104,448.826 to 77,902.286, about a 25% improvement. RMSE is 107,281.902 and R2 is -0.845. The pipeline is functional, but the short dated Amazon source prevents a production-accuracy claim.

2. Role access and connected pages

Role	Customer intelligence	Forecasting	Restriction
Business Owner	Business summary and customer membership	Business revenue and export	No AI model configuration
Store Manager	Assigned-store summary	Assigned-store demand	No business revenue or customer membership
Sales Executive	Assigned customers	Own personal forecast	No business/store forecasts or export
Administrator	Full tenant access after MFA	Business, store, personal and monitoring	Store/seller selectors remain tenant-bound

Sales Executive policy decision

The original proposal denied general forecasting access to Sales Executives. The implemented project extension is a personal-only forecast. The backend always uses the authenticated seller ID, so this does not expose business-wide forecasting.

Administrator coverage

The Administrator report selector now opens business analytics, revenue, store demand, Sales Executive personal forecasts and model monitoring. Store demand requires a selected tenant store, and personal forecasting requires a selected Sales Executive.

3. Data contracts and remaining limitations

Item	Current handling	Status / next action
Parquet sale_amount	Used as the demand target and labelled source_unit.	Provider must confirm whether it is quantity, value or an index.
Demand to inventory	Exact SKU match or explicit reviewed mapping CSV only.	Current source products 117 and 300 remain unmapped; stock risk is unknown.
Revenue history	Chronological comparison and prediction intervals are implemented.	Obtain longer dated history before production use.
Model monitoring	Versions, metrics, statuses and import jobs are visible.	Drift, matured accuracy and latency telemetry are later hardening work.
Reporting	CSV and browser Print/PDF are role-controlled.	XLSX, backend PDF jobs and saved filters are later enhancements.

Validated inventory mapping contract

The optional mapping file contains source_store_id, source_product_id, store_code and product_sku. The importer rejects unknown stores, unknown SKUs, cross-store rows and conflicting mappings. MarketMind does not create a false relationship between unrelated datasets.

Evaluation statement

Milestone 2 is ready for a functional project evaluation. Customer segmentation, behaviour analysis and all four forecasting experiences use live database records and backend-enforced role scopes. The limitations above should be stated openly as data and production-hardening work.